

OSSP-PRACTICAL-1

- Created a file named `prac1.c` using `nano`

```
vida@LAPTOP-4B7UUAUH:~$ cd os_lab
vida@LAPTOP-4B7UUAUH:~/os_lab$ nano prac1.c
```

- Compiled and ran the program where the command “`ls`” is entered into the program.

```
vida@LAPTOP-4B7UUAUH:~/os_lab$ gcc prac1.c
vida@LAPTOP-4B7UUAUH:~/os_lab$ ./a.out
ls
ls
```

- The Child PID has been returned followed by its command execution while the parent process waited till the child process completes.

```
vida@LAPTOP-4B7UUAUH:~/os_lab$ nano prac1.c
vida@LAPTOP-4B7UUAUH:~/os_lab$ gcc prac1.c
vida@LAPTOP-4B7UUAUH:~/os_lab$ ./a.out
ls
ls
Child PID is = 1284
a.out      execlp2.c  execv.c    pipe7.c    pipeextask2.c
exec3.c    execline.c newfile1    pipe7.c.save pipefork.c.save
execl1.c   exectask.c pipe6.c     pipeextask.c prac1.c
Parent PID is = 1119
```

Used uname and lscpu commands :

```
vida@LAPTOP-4B7UUAUH: ~$ uname  
Linux  
vida@LAPTOP-4B7UUAUH:~$ lscpu  
Architecture:                x86_64  
CPU op-mode(s):              32-bit, 64-bit  
Address sizes:               39 bits physical, 48 bits virtual  
Byte Order:                  Little Endian  
CPU(s):                      12  
On-line CPU(s) list:        0-11  
Vendor ID:                   GenuineIntel  
Model name:                  13th Gen Intel(R) Core(TM) i5-13420H  
  
CPU family:                  6  
Model:                       186  
Thread(s) per core:         2  
Core(s) per socket:         6  
Socket(s):                   1  
Stepping:                    2  
BogoMIPS:                    5222.40  
Flags:                        fpu vme de pse tsc msr pae mce cx8  
apic sep mtrr pge mca cmov pat pse3  
6 clflush mmx fxsr sse sse2 ss ht s  
yscall nx pdpe1gb rdtscp lm constan  
t_tsc rep_good nopl xtopology tsc_r  
eliable nonstop_tsc cpuid tsc_known  
_freq pni pclmulqdq vmx ssse3 fma c  
xl6 pcid sse4_1 sse4_2 x2apic movbe  
popcnt tsc_deadline_timer aes xsav  
e avx f16c rdrand hypervisor lahf_l  
m abm 3dnowprefetch ssbd ibrs ibpb  
stibp ibrs_enhanced tpr_shadow ept  
vpid ept_ad fsgsbase tsc_adjust bmi  
1 avx2 smep bmi2 erms invpcid rdsee  
d adx smap clflushopt clwb sha_ni x  
saveopt xsavec xgetbv1 xsaves avx_v  
nni vnmi umip waitpkg gfni vaes vpc  
lmulqdq rdpid movdiri movdir64b fsr
```

Virtualization features:

Virtualization:	VT-x
Hypervisor vendor:	Microsoft
Virtualization type:	full

Caches (sum of all):

L1d:	288 KiB (6 instances)
L1i:	192 KiB (6 instances)
L2:	7.5 MiB (6 instances)
L3:	12 MiB (1 instance)

NUMA:

NUMA node(s):	1
NUMA node0 CPU(s):	0-11

Vulnerabilities:

Gather data sampling:	Not affected
Ghostwrite:	Not affected
Indirect target selection:	Not affected
Itlb multihit:	Not affected
L1tf:	Not affected
Mds:	Not affected
Meltdown:	Not affected
Mmio stale data:	Not affected
Old microcode:	Not affected
Reg file data sampling:	Mitigation; Clear Register File
Retbleed:	Mitigation; Enhanced IBRS
Spec rstack overflow:	Not affected
Spec store bypass:	Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1:	Mitigation; usercopy/swaps barrier and __user pointer sanitization
Spectre v2:	Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRBS-eIBRS S W sequence; BHI BHI_DIS_S
Srbds:	Not affected
Tsa:	Not affected
Tsx async abort:	Not affected
Vmscape:	Not affected

arch_capabilities

Lsblk and ps commands :

```
vida@LAPTOP-4B7UUAUH:~$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk
sdb   8:16   0 159.4M  1 disk
sdc   8:32   0    2G   0 disk [SWAP]
sdd   8:48   0    1T   0 disk /mnt/wslg/distro
/
```

```
vida@LAPTOP-4B7UUAUH:~$ ps
  PID TTY          TIME CMD
  317 pts/0    00:00:00 bash
  481 pts/0    00:00:00 ps
```

```
vida@LAPTOP-4B7UUAUH:~$ top
top - 17:01:29 up 1:14, 1 user, load average: 0.04, 0.05, 0.01
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.1 us, 0.0 sy, 0.0 ni, 99.9 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7788.6 total, 7218.7 free, 555.5 used, 165.5 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7233.1 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	24320	15364	11608	S	0.0	0.2	0:00.61	systemd
2	root	20	0	3180	2208	2072	S	0.0	0.0	0:00.00	init-systemd(Ub
7	root	20	0	3180	2040	1940	S	0.0	0.0	0:00.00	init
47	root	19	-1	42200	16324	15052	S	0.0	0.2	0:00.14	systemd-journal
79	systemd+	20	0	22540	14600	11980	S	0.0	0.2	0:00.06	systemd-resolve
85	root	20	0	35428	12388	9228	S	0.0	0.2	0:00.22	systemd-udev
122	root	20	0	2888	1948	1820	S	0.0	0.0	0:00.02	chronyd-starter
131	root	20	0	4472	3164	2916	S	0.0	0.0	0:00.00	cron
142	message+	20	0	8816	5348	4676	S	0.0	0.1	0:00.04	dbus-daemon
153	root	20	0	44092	29700	14560	S	0.0	0.4	0:00.14	networkd-dispat
157	root	20	0	18436	9444	8196	S	0.0	0.1	0:00.06	systemd-logind
160	root	20	0	1792300	14184	11512	S	0.0	0.2	0:00.07	wsl-pro-service
166	syslog	20	0	220548	5596	4660	S	0.0	0.1	0:00.04	rsyslogd
213	_chrony	20	0	20420	6492	5716	S	0.0	0.1	0:00.04	chronyd
222	_chrony	20	0	12092	2456	1816	S	0.0	0.0	0:00.00	chronyd
228	root	20	0	123456	32720	18108	S	0.0	0.4	0:00.10	unattended-upgr